ELC Activity (ODD SEM 25-26)

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1. Background

1.1. Understanding ADHD: ADHD is a neurodevelopmental disorder common in children, with ongoing patterns of inattention, impulsivity, and hyperactivity. The behaviors can lead to poor grades, problems with relationships, and lowered general well-being. ADHD is thought to occur in about 11% of children worldwide, with greater frequency in boys. Early diagnosis and timely treatment are crucial for enhancing life conditions. Although accepted in prominent diagnostic systems such as DSM-5, the use of clinical judgment brings in variability as well as treatment delay. This project seeks to investigate the viability of employing EEG-based machine learning models for objective ADHD detection.

1.2. EEG-Based Detection of ADHD: Electroencephalography (EEG) provides a cost-effective and non-invasive means of measuring brain function at high temporal resolution. Research demonstrates unmistakable EEG activity in children with ADHD, most notably increased theta waves and altered theta/beta ratios. Such biological markers are utilized in this project to develop computerized diagnostic models based on EEG-based data.

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2. Review of Prior Work

2.1. Historical Context of EEG Use in ADHD: The value of EEG in psychiatric and behavioral evaluation has been acknowledged for many years. Jasper et al.'s 1938 study was one of the first to note EEG abnormalities in children with behavioral difficulties. EEG analyses since then—morphological, temporal, and frequency-based approaches—have been heavily utilized to differentiate ADHD. The development of artificial intelligence has furthered the prospects of accurate, objective detection.

2.2. Present Techniques and Limitations: Present machine learning methods integrate heterogeneous EEG features to establish prediction models. Results tend to be inconsistent because of small sample sizes, redundant features, and overfitting of models. Major limitations include inconsistent methods, poor cross-population validation, and determining strong, clinically relevant features.

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3. Identified Research Needs

3.1. Lack of Clear and Actionable Indicators: Modern methods tend to employ excessively large feature sets with no clinical verification, which complicates interpretation. There is still a necessity to identify a sparse set of features with high diagnostic utility.

3.2. Enhancing Model Transferability: Training data does not necessarily equate to high accuracy in real-world settings because of overfitting and poor generalizability. Better validation practices, including testing on external datasets, are necessary.

3.3. Exploring Temporal Signal Properties: Though EEG data is temporal by nature, most models ignore its dynamic aspects. The use of time-sensitive methods or inter-channel inter-dependencies would further augment diagnostic power.

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4. Project Objectives

4.1. Deriving Important EEG-Based Features: This research emphasizes the detection of vital morphological and statistical characteristics from EEG signals that can distinguish ADHD instances credibly and maintain a feature space parsimonious and effective.

4.2. Constructing Optimal Predictive Models: The goal encompasses creating classifiers such as SVM, MLP,Random forest that provide maximum accuracy when separating ADHD from control samples. Significance is on attentive tuning and cross-validation.

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5. Implementation Plan

5.1 APPROACH A - MACHINE LEARNING

5.1.1. Dataset Description and Cleaning: The research utilizes EEG recordings of 121 children (7–12 years old), divided into 61 ADHD cases and 60 controls. Signals were recorded from 19 scalp electrodes while performing visual attention tasks. Normalization and artifact filtering are part of the preprocessing.

5.1.2. Feature Extraction: The work obtains four sets of features:

1. Morphological Features:
These capture the signal's shape and structural characteristics:

- **Amplitude**: Indicates the peak signal strength.
- **Range**: The difference between the maximum and minimum signal values.

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2. Time-Domain Features:
These describe the statistical behavior of the signal over time:

- **Mean**: Average signal value, showing overall level.
- **Variance**: Reflects how much the signal fluctuates.
- **Skewness**: Indicates the asymmetry of the signal distribution.
- **Kurtosis**: Shows whether the signal has sharp peaks or flatness.
- **Hjorth Parameters**: Particularly *Activity* and *Mobility*, which measure signal power and complexity.

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3. Frequency-Domain Features:
These focus on the signal's spectral content:

- **Delta Band Power (0.5–4 Hz):** Often associated with deep sleep and low brain activity.
- **Theta Band Power (4–8 Hz):** Linked to drowsiness, inattention, or early cognitive processing.
- **Alpha Band Power (8–13 Hz):** Reflects relaxed but alert mental states and is often prominent during calm wakefulness.
- **Beta Band Power (13–30 Hz):** Associated with active thinking, focus, and motor behavior.
- **Total Power:** The overall signal energy across all frequency bands, giving a global measure of brain activity.
- **Relative Delta Power:** The proportion of delta power to the total power, useful for assessing dominance of low-frequency activity.
- **Relative Theta Power:** Highlights the relative contribution of theta activity, often increased in ADHD or fatigued states.
- **Relative Alpha Power:** Indicates the strength of alpha rhythms relative to total power, reflecting alert relaxation.

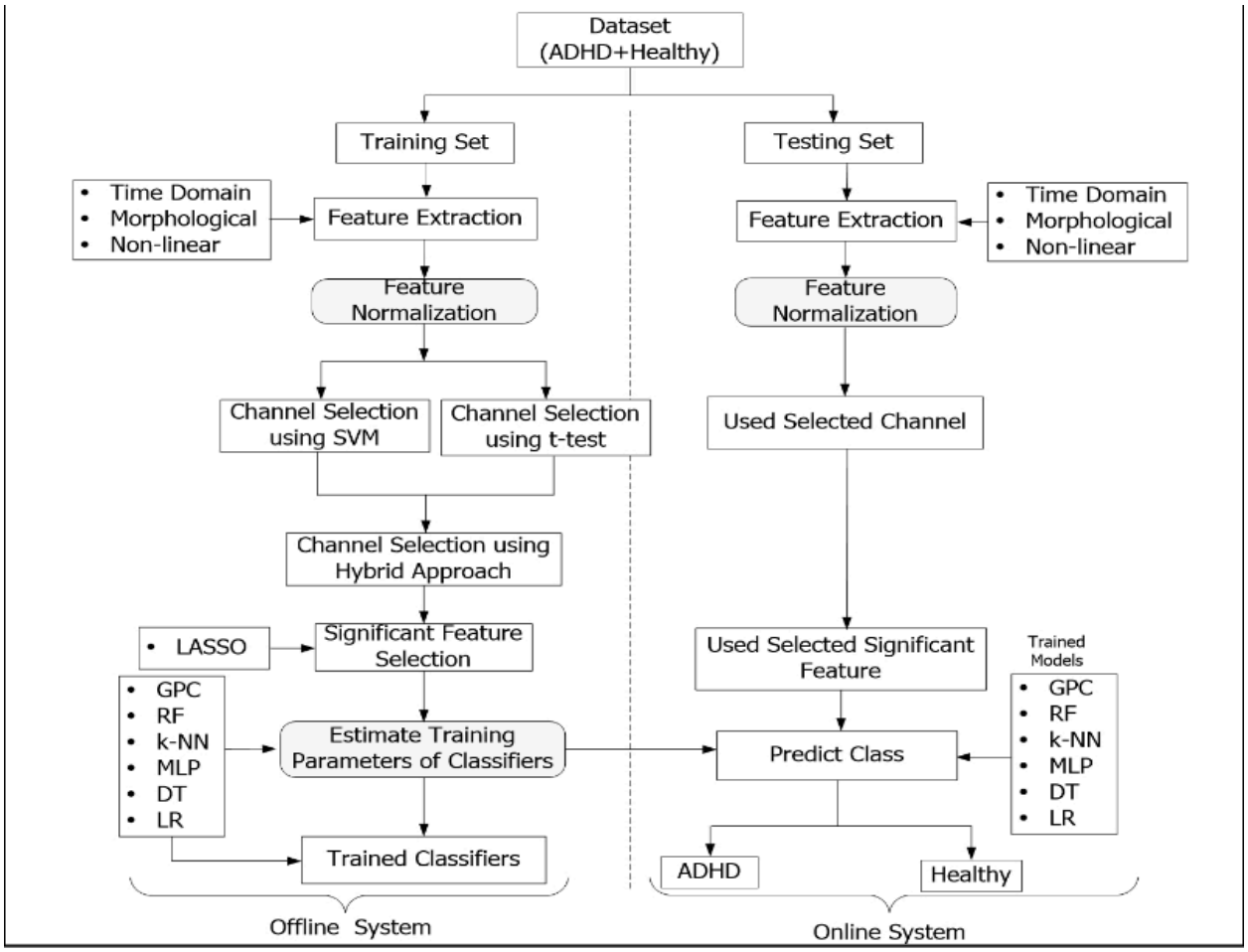
- **Relative Beta Power:** Shows the proportion of beta activity, often linked to engagement and mental workload.
- **Spectral Entropy:** Captures the irregularity and complexity of the signal's frequency distribution; higher values indicate more disordered activity.
- **Peak Frequency:** The frequency at which the signal's power is highest, often used as an individual alpha frequency marker.
- **Spectral Centroid:** The "center of mass" of the power spectrum, representing the average frequency weighted by power—helps identify shifts in spectral balance.

4. Connectivity Features:
These features evaluate the statistical and phase-based relationships between signals from different brain regions:

- **Correlation (Mean, Std, Max, Min):** Measures the linear relationship between EEG channels. Higher values indicate more synchronized activity.
- **Spearman Correlation (Mean, Std):** Captures monotonic (rank-based) relationships between signals, robust to non-linear dynamics.
- **Mutual Information (Mean, Std):** Quantifies shared information between signals, reflecting both linear and non-linear dependencies.
- **Phase Coherence (Mean, Std, Max):** Assesses the consistency of phase differences across signals, indicating phase locking or synchronization.
- **Spectral Coherence (Mean, Std):** Evaluates the consistency of frequency components across signals, useful for identifying frequency-specific coupling.
- **Cross-Correlation (Mean, Max):** Detects time-lagged similarity between EEG signals, highlighting temporal relationships across regions.
- **Cosine Similarity (Mean, Std):** Measures signal alignment in multi-dimensional space, providing a geometric view of signal similarity.
- **Phase Locking Value (PLV Mean, Std):** Quantifies phase synchronization across trials, with higher values indicating stable phase coupling.
- **Alpha Coherence (Mean):** Focuses on coherence specifically in the alpha band (8–13 Hz), often linked to attention and cognitive states.

5.1.3. Classifier Training and Testing: Four classifiers (SVM with various kernels, k-NN, MLP, and logistic regression) will be trained on the optimized feature sets. Hyperparameter tuning will be done using grid search and performance validation using LOOCV. Metrics for evaluation are accuracy, recall, specificity, F1-score, and ROC AUC.

5.1.4. Comparative Analysis: The solution will be tested relative to prior work to quantify gains and discover continued issues. This will inform subsequent best practices in EEG-based screening for ADHD.



5.2 APPROACH B - DEEP LEARNING

5.2.1. Data Acquisition and Preprocessing:

- EEG data were recorded from 19 channels at a sampling frequency of 128 Hz during visual stimulus exposure.
- Only EEG recordings with more than 9250 samples were included in the analysis.
- Signals were processed using Robust Local Mode Decomposition (RLMD) and Variational Mode Decomposition (VMD) to produce three subband signals.
- Processed signals were converted into image-like matrices for input into the deep learning model.

5.2.2. CNN Architecture:

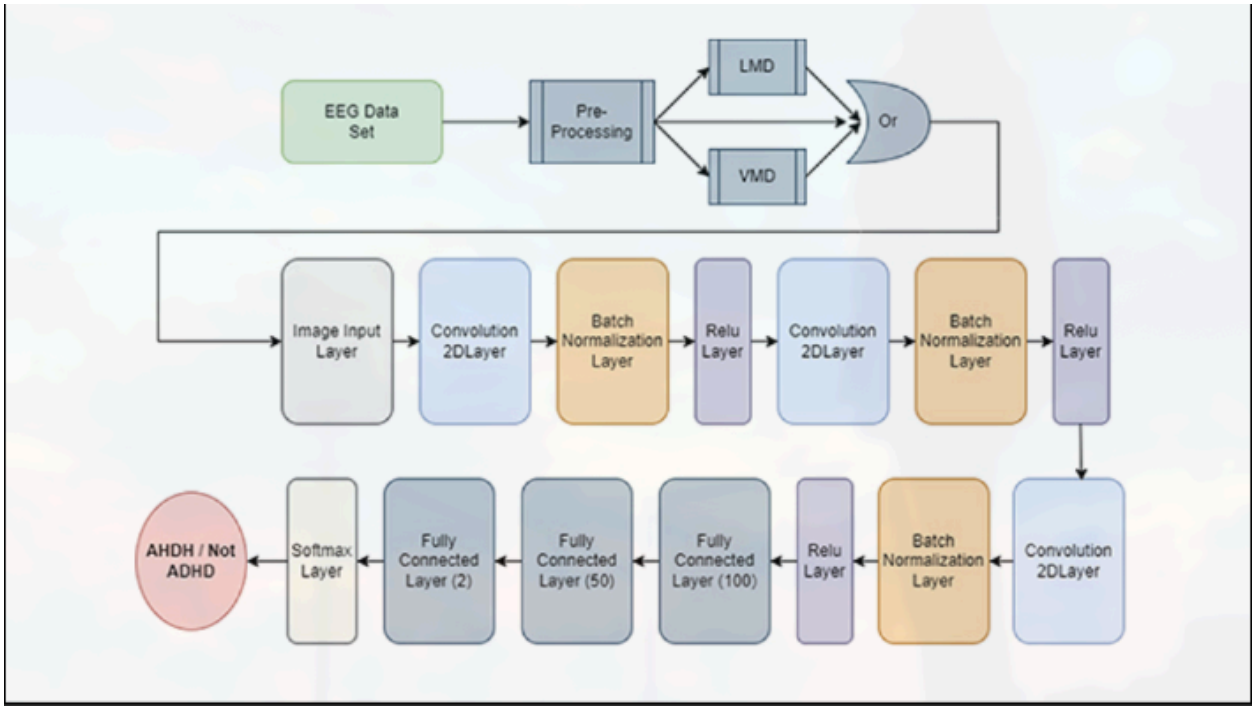
- The Convolutional Neural Network (CNN) consisted of repeated blocks of convolutional, batch normalization, and ReLU layers (applied three times).
- Followed by three fully connected layers with sizes 100, 50, and 2.
- The final classification layer used a softmax activation.

5.2.3. Model Training and Evaluation:

- Training was limited to eight epochs to balance training time and performance.
- Evaluation was performed using five-fold cross-validation.
- Confusion matrices and training durations were recorded for each fold.

5.2.4. Comparative Analysis:

- ADHD classification using all 19 EEG channels achieved:
    - 95.24% accuracy
    - 97% sensitivity
    - 94% specificity
    - 1.78% standard deviation
    - 1421 seconds training time
- Classification using only seven frontal EEG channels was achieved:
    - 94.76% accuracy
    - 96% sensitivity
    - 94% specificity
    - 2.81% standard deviation
    - 543 seconds training time



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6. Conclusion

This project illustrates how, together, accurate feature selection and sophisticated machine learning and deep learning can greatly enhance ADHD classification using EEG data. The systematic validation process adds scientific rigor. Future work involves investigating deep models and deploying real-time systems for clinical use.

Study Table

Study	Sample Size	features	Classification	Accuracy
Maniruzzaman et al. [2023]	121 children (61 ADHD, 60 Control)	28 (selected from time-domain, morphological, non-linear	Gaussian Process Classification (GPC)	96.53%
Esas and Latifoğlu et al. [2023]	120 children (61 ADHD, 59 Control)	19-channel raw EEG signals	Convolutional Neural Network (CNN)	95.24%
Esas and Latifoğlu et al. [2023]	120 children (61 ADHD, 59 Control)	7 frontal channels (Fp1, Fp2, F3, F4, F7, F8, Fz) raw EEG	Convolutional Neural Network (CNN)	94.76%
Esas and Latifoğlu et al. [2023]	120 children (61 ADHD, 59 Control)	Fp1 + 3 RLMD + 3 VMD subbands (7 features total)	Convolutional Neural Network (CNN)	87.38%
Maniruzzaman et al. [2022]	121 children (61 ADHD, 60 Control)	47 (selected via LASSO from 342 morphological and time-domain EEG features)	Support Vector Machine (SVM)	94.2%
Maniruzzaman et al. [2022]	121 children (61 ADHD, 60 Control)	47 (LASSO-selected EEG features)	Multilayer Perceptron (MLP)	93.4%
Li and Guo et al. [2025]	57 children (29 ADHD, 28 Control)	19-channel resting EEG; LSTM (time), CNN (frequency), PLI + COH fused functional connectivity	Graph Convolutional Network (GCN)	97.29%
Li and Guo et al. [2025]	121 children	19-channel task	Graph	96.67%

Study Table

Study	Sample Size	features	Classification	Accuracy
	(61 ADHD, 60 Control)	EEG; LSTM (time), CNN (frequency), PLI + COH fused functional connectivity	Convolutional Network (GCN)	

ML Model Accuracy

Model	Training Accuracy	Testing Accuracy
GPC	98.9	92.6
Random Forest	100	95.8
SVM	100	95.83
MLP	98	87.8

DL Model Accuracy

Model	Training Accuracy	Testing Accuracy
CNN+LSTM+vmd decompositon	99.21	91.14
LSTM	99.33	97.58
BiLSTM	99.79	97.64
CNN+vae features	98.50	98.60

Feature Extraction:

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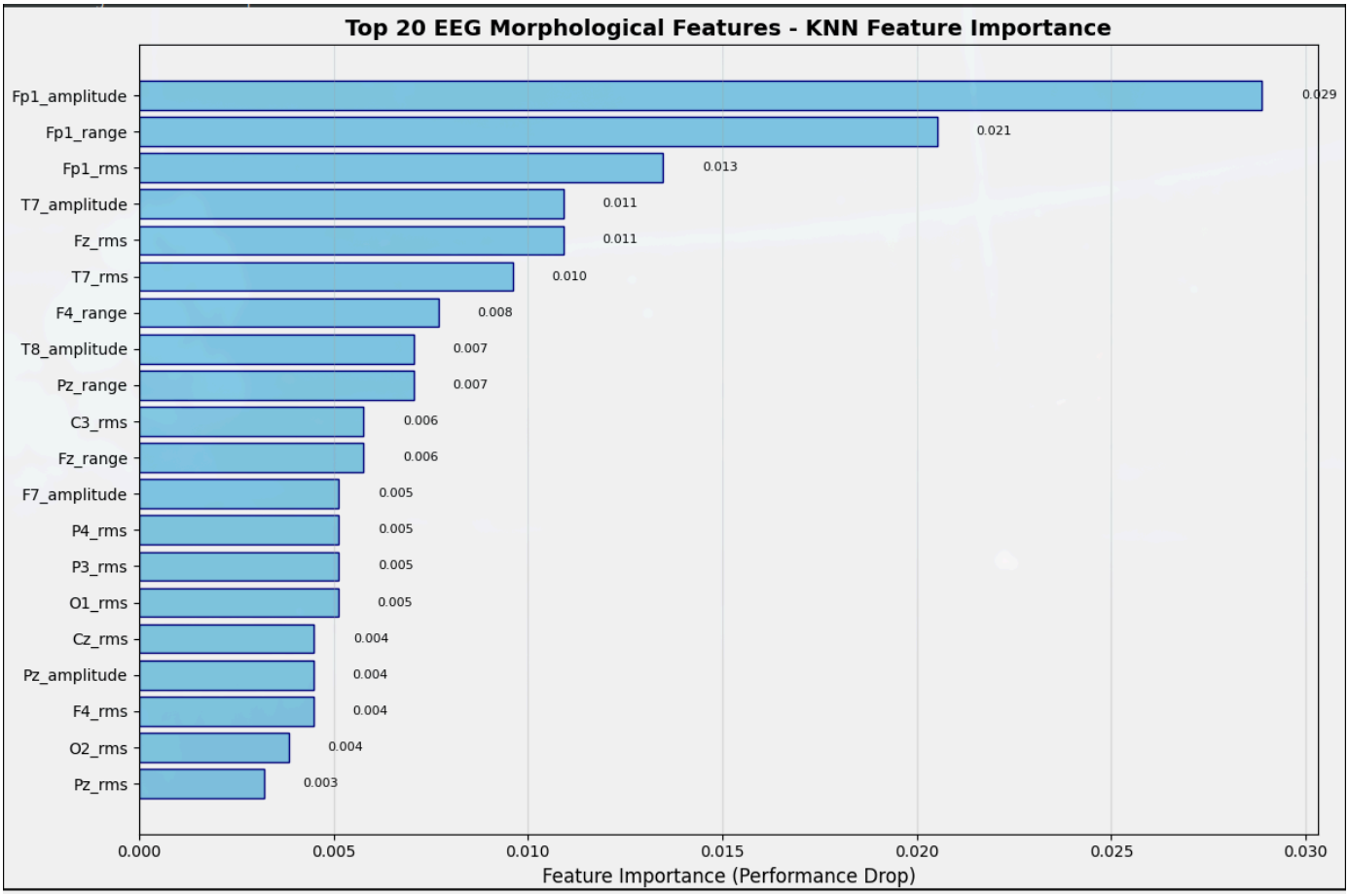
1. Morphological Features

We used 3 morphological features across 19 EEG channels, resulting in a total of 57 morphological features.

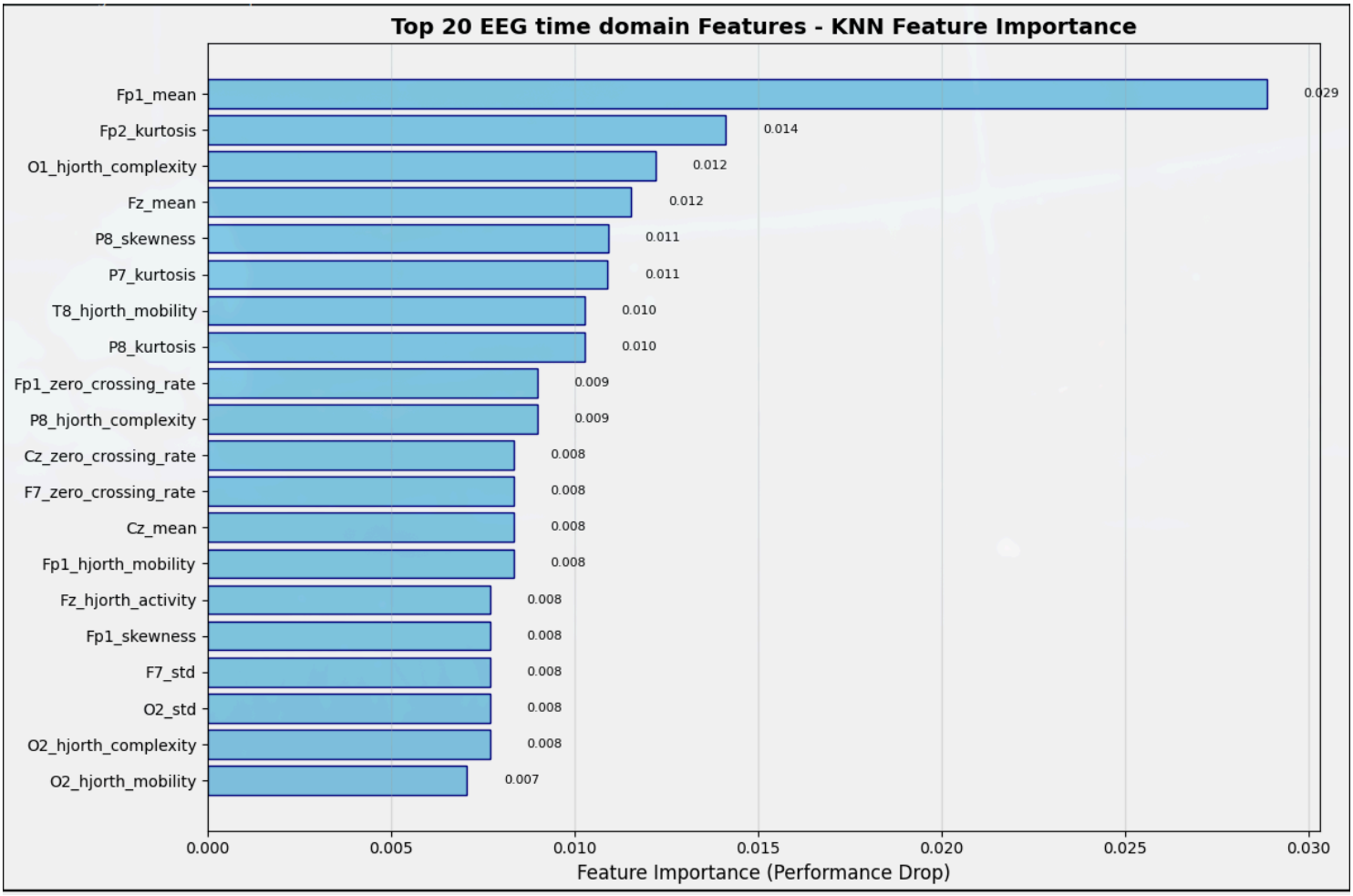
Morphological features
Amplitude
Range
Rms



2. Time Domain Features

We used 5 Time Domain features across 19 EEG channels, resulting in a total of 95 Time Domain features.

Time Domain Features
Mean
Variance
Skewness
Kurtosis
Hjorth parameters

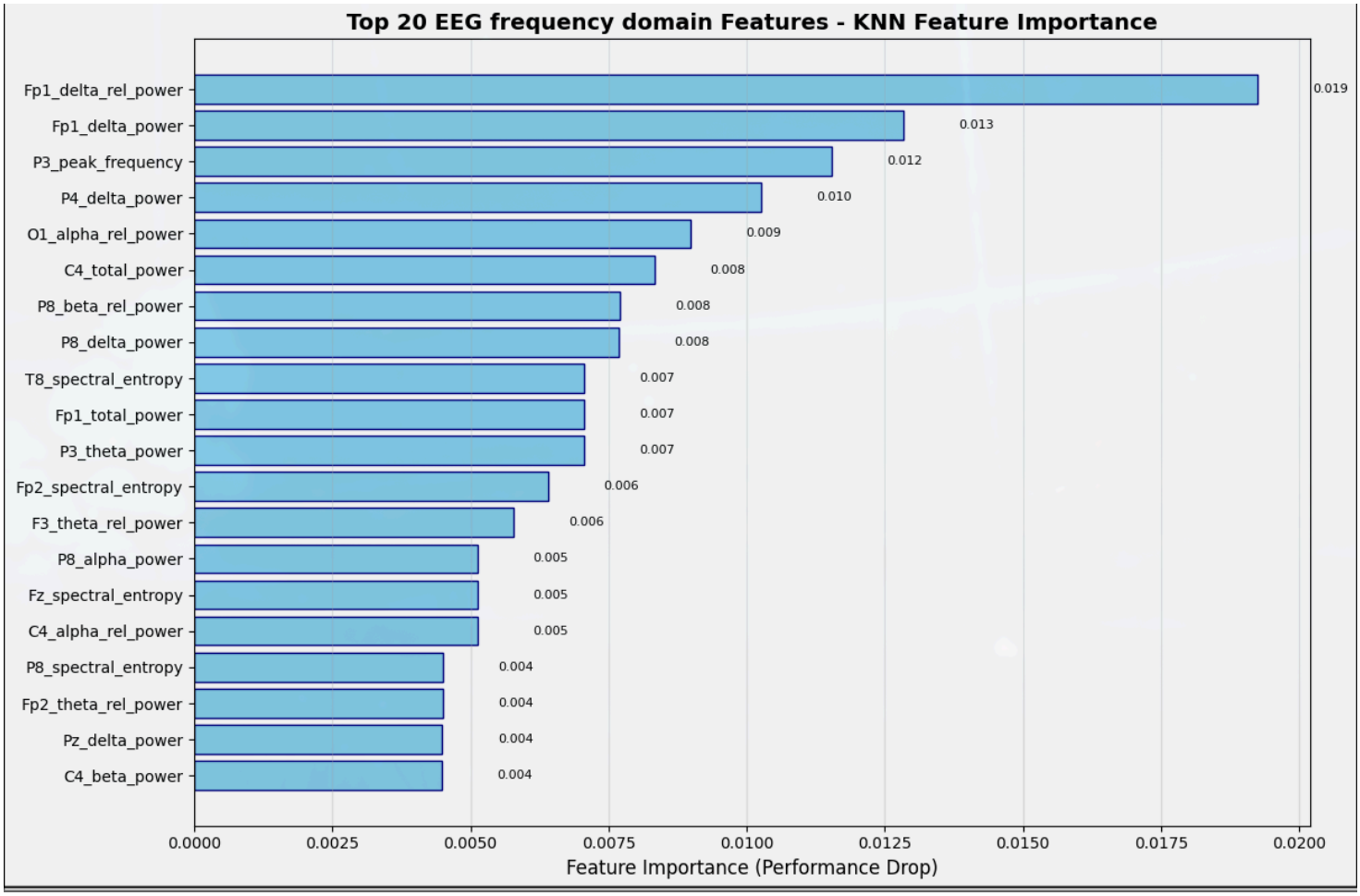


3. Frequency Domain Features

We used 12 Frequency Domain features across 19 EEG channels, resulting in a total of 228 Frequency Domain features.

Frequency Domain Features
Delta Band Power
Theta Band Power
Beta Band Power
Alpha Band Power
Total power

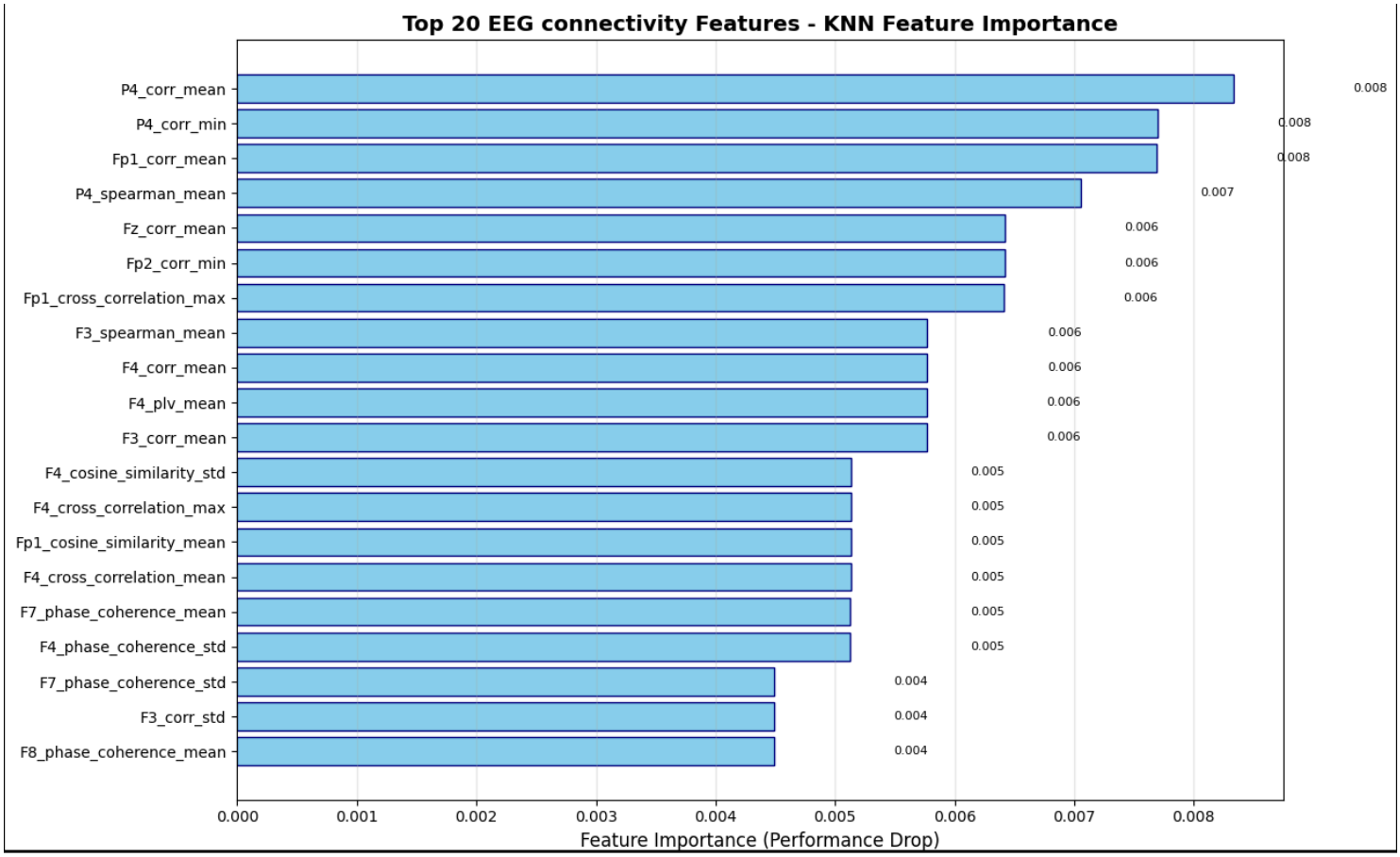
Frequency Domain Features
Relative Delta Power
Relative Theta Power
Relative Beta Power
Relative Alpha Power
Special Entropy
Peak Frequency
Special Centroid



4. Connectivity Features :-

We used 20 Connectivity Domain features across 19 EEG channels, resulting in a total of 380 Frequency Domain features.

Connectivity Features
Correlation (Mean,Std,Max,Min)
Spearman Correlation (Mean,Std)
Mutual Information (Mean,Std)
Phase Coherence (Mean,Std,Max)
Spectral Coherence (Mean,Std)
Cross-Correlation (Mean,Max)
Cosine Similarity (Mean, Std)
Phase locking Value (PLV Mean, Std)
Alpha Coherence (Mean)


Top 20 EEG connectivity Features - KNN Feature Importance
P4_corr_mean
P4_corr_min
Fp1_corr_mean
P4_spearman_mean
Fz_corr_mean
Fp2_corr_min
Fp1_cross_correlation_max
F3_spearman_mean
F4_corr_mean
F4_plv_mean
F3_corr_mean
F4_cosine_similarity_std
F4_cross_correlation_max
Fp1_cosine_similarity_mean
F4_cross_correlation_mean
F7_phase_coherence_mean
F4_phase_coherence_std
F7_phase_coherence_std
F3_corr_std
F8_phase_coherence_mean
0.008
0.008
0.008
0.007
0.006
0.006
0.006
0.006
0.006
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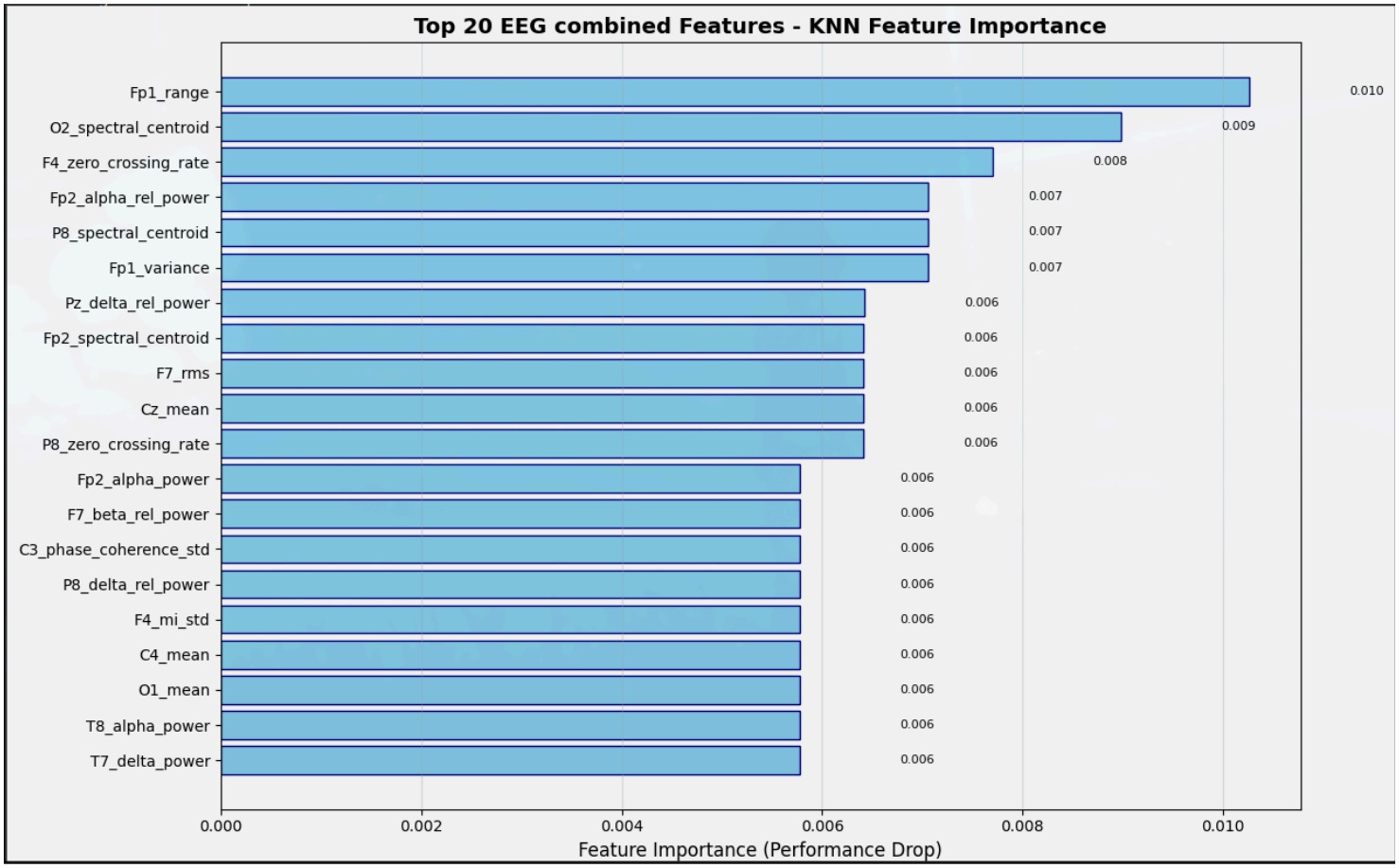
0.004
0.004
0.000
0.001
0.002
0.003
0.004
0.005
0.006
0.007
0.008
Feature Importance (Performance Drop)

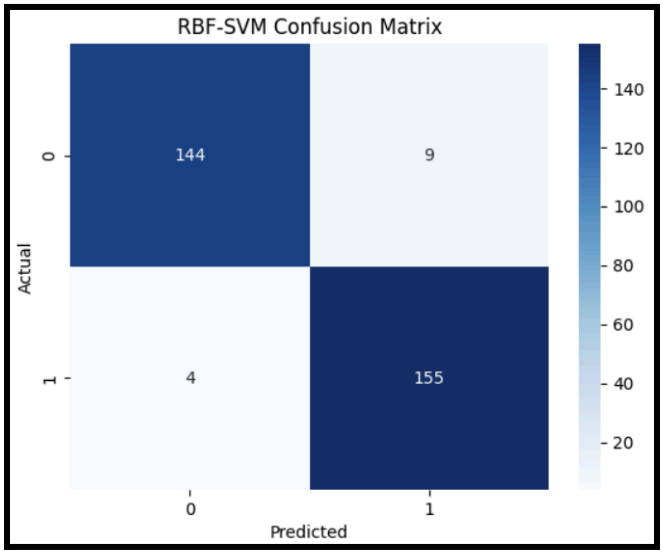
Total Combined Features :-

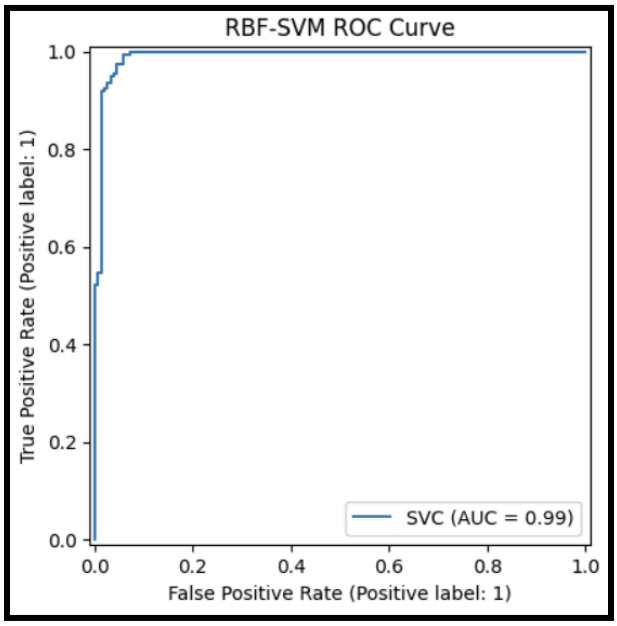
Combining all these features , we got a total of 760 features.



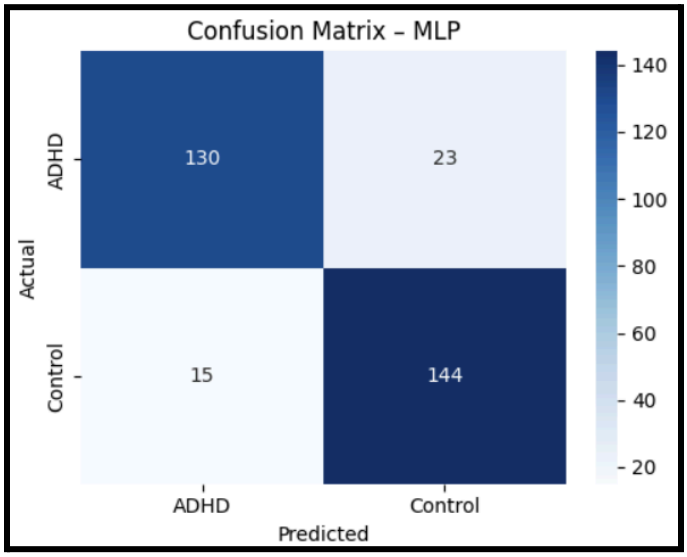
Performance:

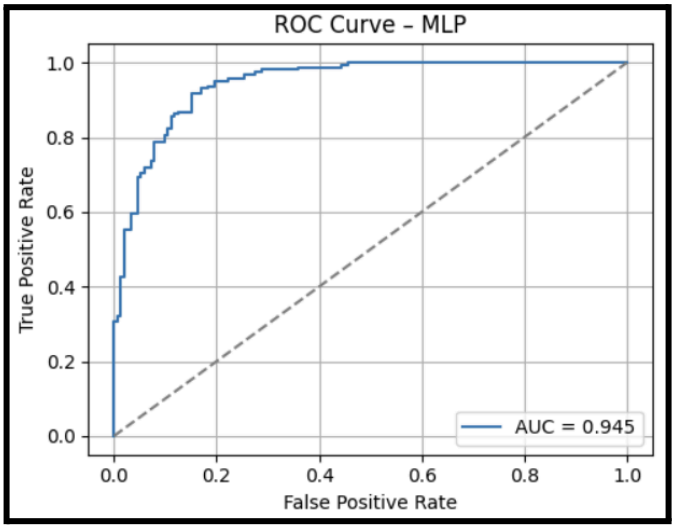
1) SVM



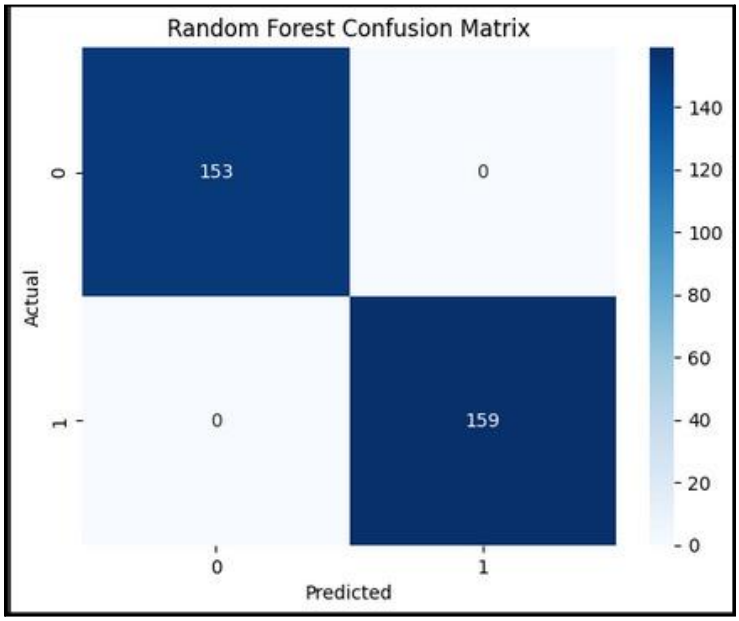


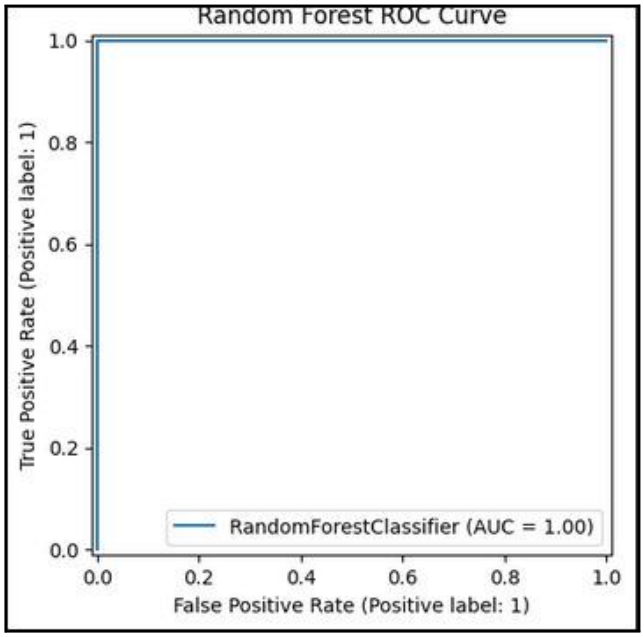
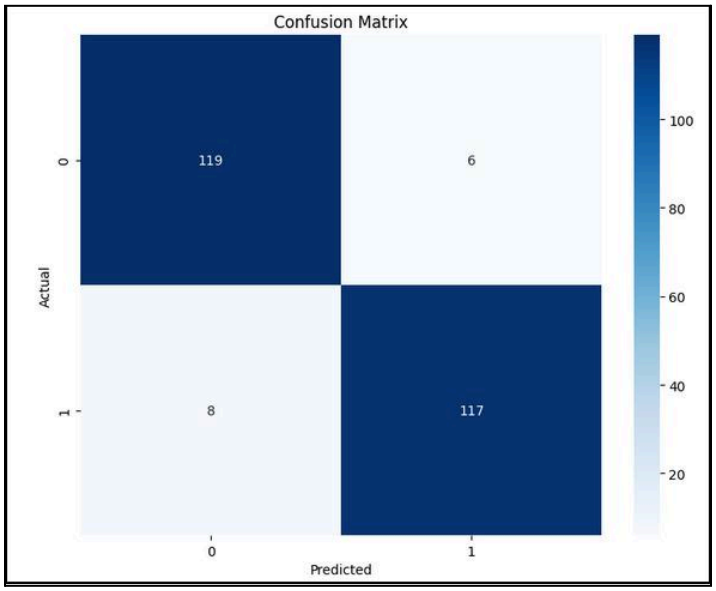
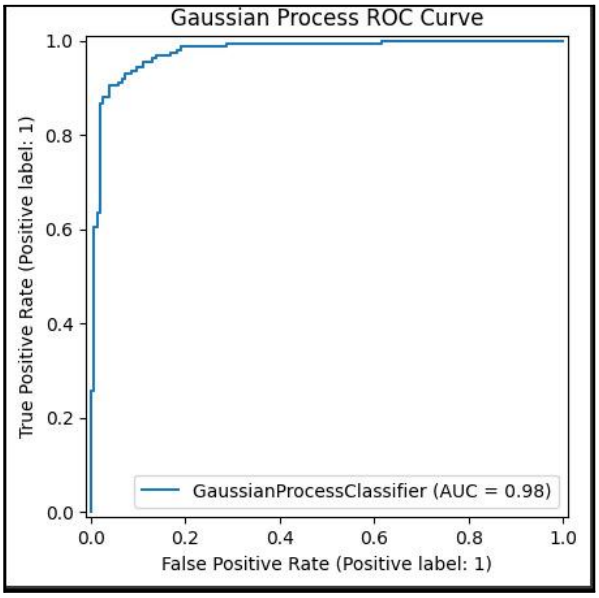
2) MLP





3) Random Forest





4) GPC